

Market-Implied Credit Stress Beyond the Regulatory Vasicek Framework

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1 Introduction

The Basel regulatory Vasicek framework measures credit risk through a single systematic factor assumed to follow a standard normal distribution. Regulatory capital is obtained by evaluating the model at a fixed confidence level, typically 99.9 per cent, corresponding to an extreme but predetermined credit stress scenario. This note considers an empirical complement to the regulatory Vasicek framework based on observed credit market data. Rather than relying solely on a theoretical Gaussian factor, the approach extracts two common credit factors from corporate bond spread indices across regions and rating categories. These factors are combined into a single measure of prevailing credit conditions and used to estimate time-varying portfolio loss rates within a Vasicek framework.

The objective is not to replace the regulatory Vasicek model, which serves an important role as a transparent and consistent benchmark for capital assessment. Instead, the purpose is to examine whether market-based measures of credit conditions can provide complementary information about systemic credit risk and to explore some of the key assumptions underlying the regulatory specification. In particular, the analysis focuses on the behaviour of the empirical credit factor during periods of market stress and on the extent to which observed credit market outcomes are consistent with the Gaussian assumption embedded in the regulatory framework. By comparing market-implied credit states with the fixed regulatory stress scenario, the exercise provides a simple empirical perspective on the strengths and limitations of the standard normality assumption.

2 Regulatory Vasicek Model

2.1 Asset Value Representation

Consider an obligor whose latent asset return is represented by

$$A_i = \sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i, \tag{1}$$

where $Z \sim N(0, 1)$ is the systematic credit factor, $\varepsilon_i \sim N(0, 1)$ is an idiosyncratic factor and ρ denotes the asset correlation parameter. Default occurs whenever

$$A_i < \Phi^{-1}(PD), \quad (2)$$

where PD denotes the unconditional probability of default and $\Phi(\cdot)$ is the standard normal cumulative distribution function. The conditional probability of default given a realization of the systematic factor Z is

$$\theta(Z) = \Phi\left(\frac{\Phi^{-1}(PD) - \sqrt{\rho}Z}{\sqrt{1-\rho}}\right). \quad (3)$$

2.2 Regulatory Stress Scenario

Under the Basel IRB framework the systematic factor is assumed to take its α -quantile realization

$$Z_\alpha = \Phi^{-1}(\alpha), \quad (4)$$

where typically

$$\alpha = 99.9\%. \quad (5)$$

The resulting stressed conditional default probability is

$$\theta_{\text{reg}} = \Phi\left(\frac{\Phi^{-1}(PD) - \sqrt{\rho}\Phi^{-1}(\alpha)}{\sqrt{1-\rho}}\right). \quad (6)$$

This formulation assumes that the credit state is entirely determined by a theoretical normal random variable. No information from observed credit markets enters the model.

3 Multi-Factor Vasicek Framework

3.1 Motivation

The assumption that the systematic credit factor follows a single standard normal variable is attractive for regulatory capital calculation but less suitable for monitoring actual credit conditions. Credit markets are influenced by several common drivers including

- overall credit risk appetite,
- economic growth expectations,
- funding conditions,
- market liquidity,
- sector-specific stress.

Consequently, the latent credit state may be better represented by a combination of several systematic factors.

3.2 General Specification

A multifactor Vasicek specification can be written as

$$A_i = \sum_{k=1}^K \beta_k F_k + \sqrt{1 - \sum_{k=1}^K \beta_k^2} \varepsilon_i, \quad (7)$$

where F_k are systematic factors, β_k are factor loadings and ε_i is idiosyncratic risk. The standard single-factor model becomes a special case when $K = 1$.

3.3 From Multiple Factors to a Credit State Variable

Consider the two-factor specification

$$A_i = \beta_1 F_1 + \beta_2 F_2 + \sqrt{1 - \beta_1^2 - \beta_2^2} \varepsilon_i, \quad (8)$$

where

$$F_1, F_2, \varepsilon_i \sim N(0, 1) \quad (9)$$

and the factors are assumed independent. Conditional on a realization of the systematic factors, the portfolio loss rate is

$$\theta(F_1, F_2) = \Phi \left(\frac{\Phi^{-1}(PD) - \beta_1 F_1 - \beta_2 F_2}{\sqrt{1 - \beta_1^2 - \beta_2^2}} \right). \quad (10)$$

The credit environment may be viewed as a point in a two-dimensional factor space described by

$$(F_{1,t}, F_{2,t}). \quad (11)$$

However, portfolio losses depend only on the linear combination

$$Y_t = \beta_1 F_{1,t} + \beta_2 F_{2,t}. \quad (12)$$

Different factor realizations may therefore generate identical portfolio losses. For example, if $\beta_1 = \beta_2 = 0.5$, then $(F_1, F_2) = (4, 2)$ and $(F_1, F_2) = (3, 3)$ both produce $Y = 3$. From the perspective of portfolio credit losses these two states are therefore equivalent, even though they represent different underlying market conditions. The two-dimensional factor space is therefore projected onto a single direction determined by the loading vector

$$(\beta_1, \beta_2).$$

Since the factors are independent and Gaussian,

$$Y \sim N(0, \beta_1^2 + \beta_2^2). \quad (13)$$

Defining

$$\rho = \beta_1^2 + \beta_2^2 \quad (14)$$

and

$$Z = \frac{Y}{\sqrt{\rho}}, \quad (15)$$

it follows that

$$Z \sim N(0, 1). \quad (16)$$

The model can therefore be rewritten as

$$A_i = \sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i, \quad (17)$$

which is formally identical to the standard Vasicek specification.

At first sight this result may suggest that the multifactor formulation offers no advantage relative to the one-factor model. However, this conclusion applies only at the level of the portfolio loss representation. Economically, the interpretation remains fundamentally different. In the regulatory Vasicek framework the systematic factor is an unobservable theoretical quantity assumed to satisfy

$$Z \sim N(0, 1). \quad (18)$$

The regulatory stress scenario is obtained by setting

$$Z = \Phi^{-1}(99.9\%). \quad (19)$$

In contrast, the empirical framework derives systematic factors directly from observed credit market information. The factors may be interpreted as distinct dimensions of credit conditions, such as broad credit market stress, differences between sovereign and private credit risk, or variation between investment-grade and lower-rated credit segments. The purpose of the multifactor representation is therefore not primarily to derive a new closed-form Vasicek formula. Rather, it provides a framework for constructing an observable and market-implied measure of the prevailing credit environment.

3.4 When Does the Multifactor Structure Matter?

The previous discussion showed that a multifactor specification may be reduced to a one-factor Vasicek model when all portfolio segments share identical factor loadings. In that case, portfolio losses depend only on a single linear combination of the underlying factors. However, this does not imply that credit markets themselves are one-dimensional. Principal component analysis may indicate that multiple factors are required to explain observed credit spread dynamics. For example, the first principal component may explain only around 50 per cent of the total variation, while additional components contribute materially to the remaining dynamics. A distinction must therefore be made between the dimensionality of observed credit market data and the dimensionality of the portfolio loss model. Credit markets may be intrinsically multifactorial even when portfolio losses can be described by a single state variable. The multifactor structure becomes fully relevant when different portfolio segments exhibit different factor loadings. In that case, the portfolio loss can no longer be represented by a single common factor and the analytical Vasicek solution is no longer available.

4 Empirical Credit Factor Construction

4.1 Credit Spread Data

The empirical model uses spreads of long maturity corporate bonds relative to swap rates. The data set contains representative credit spread indices covering multiple regions and rating categories. Examples include

- United States: AAA, AA, A and BBB,
- Euro Area: AAA, AA, A and BBB,
- United Kingdom: A and BBB,
- Sweden: Mortgage Bonds.

The spreads are interpreted as market-based measures of perceived credit risk. Rising spreads indicate deteriorating credit conditions, while narrowing spreads indicate improving conditions. Figure 1 shows an example of the input data used to extract the systematic credit factor. Although spread levels differ across regions and rating categories, the series exhibit a strong common component during periods of market stress. This common variation motivates the use of principal component analysis as a tool for identifying an underlying credit state.

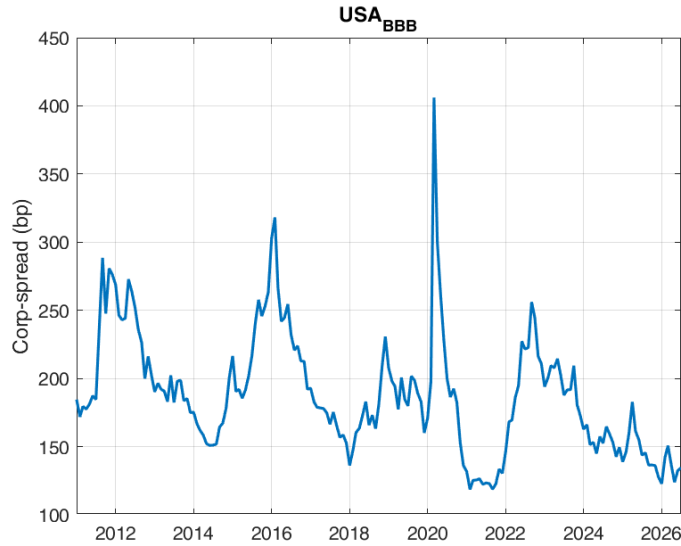


Figure 1: One example of the corporate bond spread indices used in the construction of the empirical credit factor. The full data set comprises multiple regions and rating categories and spans both benign and stressed credit market conditions.

4.2 Standardization

Each spread series is standardized according to

$$X_{i,t} = \frac{S_{i,t} - \bar{S}_i}{\sigma_i}, \quad (20)$$

where $S_{i,t}$ is the observed spread, $\bar{S}_i$ is the sample mean and σ_i is the sample standard deviation. This step removes differences in scale across rating classes and geographical markets.

4.3 Principal Component Analysis

The standardized spread matrix is used to estimate a covariance matrix from which principal components are extracted. The first two principal components explain more than 84 per cent of the total variation in the credit spread data, indicating that a small number of common factors capture the dominant dynamics of credit markets across regions and rating categories. Let

$$F_1, F_2 \quad (21)$$

denote the first two principal components and let

$$w_1, w_2 \quad (22)$$

denote their relative explained variance weights. The empirical credit factor is then defined as

$$Z_t = \sqrt{w_1}F_{1,t} + \sqrt{w_2}F_{2,t}. \quad (23)$$

Finally, the factor is standardized

$$Z_t^* = \frac{Z_t - \bar{Z}}{\sigma_Z}. \quad (24)$$

This creates a single time-varying measure of the prevailing credit environment derived from observed market prices. By construction, the factor reflects the common component driving the majority of observed movements in corporate credit spreads while filtering out a large share of idiosyncratic variation specific to individual markets and rating segments.

5 Dynamic Vasicek Conditional Loss

The empirical credit factor replaces the regulatory factor realization. The conditional portfolio loss at time t becomes

$$\theta_t = \Phi \left(\frac{\Phi^{-1}(PD) - \sqrt{\rho}Z_t^*}{\sqrt{1 - \rho}} \right). \quad (25)$$

For an asymptotically granular portfolio, the conditional default rate coincides with the conditional portfolio loss rate. The quantity θ_t may therefore be interpreted as the portfolio loss implied by the observed credit state. Unlike the regulatory model, the credit state varies through time and responds directly to developments in credit markets. Periods of systemic stress generate large positive realizations of Z_t^* and consequently elevated conditional loss rates.

6 Comparison with the Regulatory Framework

The key distinction between the two approaches concerns the treatment of the systematic credit factor. The regulatory model measures credit risk under a predefined extreme stress scenario. The empirical model instead estimates the current credit state using information embedded in corporate bond spreads.

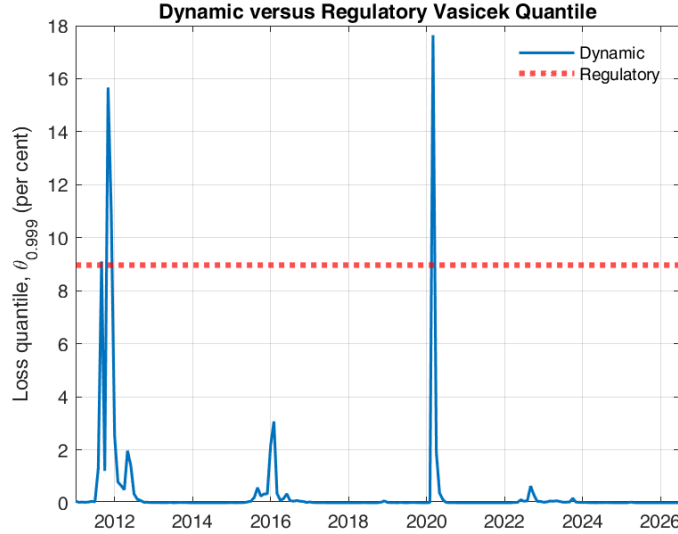


Figure 2: Dynamic Vasicek conditional loss based on the empirical credit factor compared with the fixed regulatory Vasicek loss level corresponding to a 99.9% confidence level.

Figure 2 compares the conditional loss implied by the empirical credit factor with the corresponding regulatory Vasicek quantile. During normal market conditions the empirical quantile remains well below the regulatory benchmark. However, during major episodes of credit and financial stress the empirical quantile increases sharply and temporarily exceeds the regulatory 99.9% quantile. This observation is noteworthy. The regulatory framework interprets the 99.9% confidence level as an extremely severe but plausible credit event. Yet historical market data suggest that realized credit market conditions may at times

correspond to states that are more adverse than those implied by the standard regulatory calibration. The explanation for this result is explored in the next section through an examination of the distribution of the empirical factor itself.

7 Empirical Observations

The previous section showed that the dynamic conditional loss occasionally exceeds the corresponding regulatory Vasicek loss level. Figure 3 provides a possible explanation for this result.

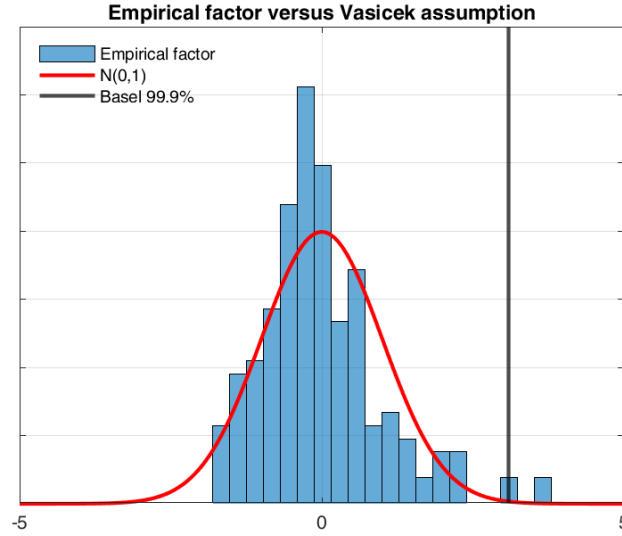


Figure 3: Distribution of the empirical credit factor compared with a standard normal distribution. The vertical line represents the Basel 99.9% stress threshold.

The empirical credit factor is approximately normal distributed over most of the sample period. However, the distribution exhibits a small number of observations in the upper tail that lie substantially beyond what would be expected under a standard normal specification. These observations coincide with major episodes of credit and financial market stress characterised by rapidly widening corporate bond spreads. As a consequence, the empirical factor occasionally reaches values that exceed the stress level associated with the Basel 99.9% confidence threshold. This explains why the dynamic Vasicek quantile can temporarily exceed its regulatory counterpart during severe market dislocations. The results suggest that realised credit market conditions may exhibit fatter tails than those implied by the Gaussian assumption underlying the regulatory Vasicek framework. While the empirical factor is broadly consistent with

normality during ordinary market conditions, extreme credit events appear to be more severe than would be implied by a standard normal distribution. This finding does not imply that the Vasicek framework is unsuitable for regulatory capital calculations. Rather, it highlights the need for caution when interpreting the Gaussian assumption as a complete description of extreme credit market outcomes. Market-based measures of credit conditions may contain information about systemic stress that is not fully reflected in the standard regulatory specification.

8 Full Multi-factor Portfolio Model

The previous sections considered a reduced-form implementation in which the first two principal components were combined into a single empirical credit factor. This approach preserves the analytical simplicity of the Vasicek framework while incorporating information from multiple segments of the credit market. An alternative approach is to retain the full multifactor structure and allow different portfolio segments to respond differently to the underlying risk factors. In this case, the systematic credit state can no longer be represented by a single common factor. Instead, portfolio losses depend on the entire vector of systematic factors and on the specific factor exposures of individual obligors or portfolio segments. The principal advantage of this approach is that it preserves potentially important information regarding the sources of credit stress. For example, sovereign-related stress and private-sector credit stress may affect different parts of a portfolio to different degrees. The resulting loss distribution therefore reflects not only the overall level of systemic stress but also its composition. The cost of this additional flexibility is that the analytical Vasicek quantile is generally no longer available. Portfolio loss quantiles must instead be obtained numerically by integrating over the joint distribution of the systematic factors.

8.1 Monte Carlo Estimation of Portfolio Loss Quantiles

Let

$$F = (F_1, F_2)' \quad (26)$$

denote the vector of systematic factors. Assuming an asymptotically granular portfolio, idiosyncratic risk is fully diversified and portfolio losses converge to their conditional expectation. Consequently, it is sufficient to simulate the systematic factors only. The portfolio loss distribution may then be estimated using Monte Carlo simulation. For each simulation run s :

1. Draw a realization of the systematic factors

$$(F_1^{(s)}, F_2^{(s)}) \quad (27)$$

from their joint distribution.

2. Compute conditional default probabilities

$$\theta_i^{(s)} = \Phi \left(\frac{\Phi^{-1}(PD_i) - \beta_{i1}F_1^{(s)} - \beta_{i2}F_2^{(s)}}{\sqrt{1 - \beta_{i1}^2 - \beta_{i2}^2}} \right). \quad (28)$$

3. Compute the corresponding portfolio loss

$$L^{(s)} = \sum_i EAD_i \cdot LGD_i \cdot \theta_i^{(s)}. \quad (29)$$

Repeating this procedure for a large number of simulations produces the empirical portfolio loss distribution

$$\{L^{(1)}, L^{(2)}, \dots, L^{(N)}\}. \quad (30)$$

The loss quantile at confidence level α is then estimated as

$$\theta_\alpha = F_L^{-1}(\alpha). \quad (31)$$

This approach preserves the multidimensional structure of the systematic credit state and allows different rating classes, sectors and geographical regions to respond differently to the same factor realization. The multifactor structure becomes economically relevant only when at least two portfolio segments exhibit different factor loadings. Otherwise, portfolio losses depend solely on a single linear combination of the underlying factors, and the model can be reduced to an equivalent one-factor Vasicek specification.

8.2 Implications for the Present Analysis

The present note does not pursue a full multi-factor portfolio specification. Instead, the first two principal components are combined into a single empirical credit state variable. This choice is deliberate. It preserves most of the variation in observed credit spreads while retaining the analytical simplicity of the Vasicek framework. The objective is therefore not to estimate portfolio loss distributions through a fully heterogeneous factor structure. Rather, the purpose is to investigate whether market-implied credit conditions behave differently from the theoretical Gaussian factor assumed in the regulatory framework. The composite factor may therefore be viewed as a dimension reduction of the underlying multifactor credit environment into a single market-implied state variable.

9 Concluding Remarks

This note presents an initial version of a dynamic Vasicek framework in which the systematic credit factor is estimated from observed corporate bond spreads rather than assumed to follow a theoretical regulatory stress scenario. The preliminary results indicate that the extracted factor captures economically meaningful periods of credit stress and, in some cases, generates loss quantiles that

exceed the regulatory 99.9% benchmark. This behaviour appears to be driven by empirical tail events that are more extreme than those implied by a standard normal distribution. The current specification should be viewed as a first prototype rather than a final model. Future work may consider alternative credit spread datasets, additional market indicators, alternative factor extraction methods, and explicit modelling of tail risk. Nevertheless, even at this early stage, the analysis highlights the importance of carefully assessing normality assumptions when modelling systematic credit risk. Historical market data suggest that extreme credit conditions can be more severe than implied by the Gaussian framework that underlies the regulatory Vasicek model.